

Bayesian LSTM Stochasticity Analysis for Sequence Modeling and Uncertainty Estimation

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Abstract

We study the impact of different stochasticity levels in Bayesian LSTM models for sequence modeling and uncertainty estimation. We compare four variants: a fully deterministic LSTM, a partially stochastic LSTM with a stochastic final layer trained via mean-field variational inference, a fully stochastic LSTM using the same inference method, and a final-layer stochastic LSTM trained with Stochastic Gradient Langevin Dynamics (SGLD). Despite the common assumption that full stochasticity yields greater expressiveness, our experiments reveal no consistent performance advantage. Partially stochastic models — especially those with stochasticity limited to the output layer — achieve comparable or superior results in both predictive accuracy and uncertainty quantification, while offering significant computational gains.

Keywords: *Bayesian LSTM, variational inference, SGLD, uncertainty quantification, stochastic neural networks, Euler angle prediction, microstructure modeling*

1. Introduction

Bayesian neural networks (BNNs) provide a principled framework for uncertainty modeling by treating parameters as distributions rather than point estimates. In sequence modeling tasks, capturing uncertainty is often as important as predictive accuracy — motivating the extension of this paradigm to recurrent architectures such as Long Short-Term Memory (LSTM) networks.

Traditionally, BNNs assume full stochasticity across all weights and biases, with approximate inference via methods such as mean-field variational inference or SGLD. However, this poses practical challenges: inference is expensive, sensitive to initialization, and may fail to accurately represent the true posterior at scale.

In this project we empirically evaluate whether full stochasticity is necessary. We compare four Bayesian LSTM variants on the task of predicting Euler angle evolution in synthetic microstructure data under incremental strain, and find that selective stochasticity in the output layer offers a better trade-off between performance and computational cost.

1.1 Problem Description

We aim to systematically compare four Bayesian LSTM variants on predicting the evolution of Euler angles in synthetic microstructure data under incremental strain:

- **Fully Deterministic LSTM** — standard point-estimate baseline
- **Partially Stochastic LSTM (Variational Output Layer)** — mean-field VI on final layer only
- **Fully Stochastic LSTM** — mean-field variational inference across all parameters
- **Partially Stochastic LSTM (SGLD Output Layer)** — Langevin dynamics on final layer only

The objective is to evaluate whether selective stochasticity can offer comparable predictive performance and uncertainty estimates with reduced computational overhead, informing the design of efficient uncertainty-aware models for time-dependent scientific prediction.

1.2 Related Work

Sharma et al. (2023) in *Do Bayesian Neural Networks Need To Be Fully Stochastic?* demonstrate both theoretically and empirically that partial stochasticity — stochasticity limited to select layers — can serve as universal conditional distribution approximators, often matching or outperforming fully stochastic models across various inference schemes. This provides a strong foundation for rethinking Bayesian deep learning with an emphasis on practicality.

Prior undergraduate research (UGP) applied a deterministic LSTM to simulate Euler angle evolution in synthetic microstructures, demonstrating strong extrapolation performance but without uncertainty estimation — a critical gap for scientific and safety-critical deployment.

1.3 Novelty

This work unifies both strands: we introduce controlled stochasticity into a high-performing LSTM architecture to bridge extrapolation accuracy with principled uncertainty quantification. We provide a systematic empirical comparison across inference schemes (VI and SGLD), bridging a gap between theoretical expressiveness and practical deployment in materials science.

2. Tools and Software

- **Python 3.9** — Core programming language for data processing, modelling, and experimentation.
- **PyTorch** — Deep learning framework for LSTM implementation and neural network training.
- **Pyro** — Probabilistic programming library built on PyTorch; used for variational inference and SGLD.
- **NumPy** — Numerical array operations and efficient data preprocessing.
- **Pandas** — Dataset manipulation and transformation into model-ready format.
- **Scikit-learn** — Model evaluation and performance metrics including MSE computation.
- **CUDA / Kaggle T4×2** — GPU acceleration for high-performance model training on Kaggle infrastructure.
- **ATEX** — Microstructure generation from Euler angles; IPF map visualisation and analysis.

3. Dataset Description

The dataset consists of **20 CSV files** — 19 used for training and 1 reserved for testing — each representing material data captured under a specific strain condition (1%–20%). Files are structured as sequential data enabling time-series modelling of material property evolution.

Features

- **Phi1, Phi2** — Crystallographic orientation angles describing the material's internal microstructure as strain increases.
- **KAM (Kernel Average Misorientation)** — Quantifies misorientation between adjacent grains; used to assess deformation and dislocation density.
- **Y, X** — Spatial coordinates of sample positions during material testing.

Training files (1–19) teach models how material properties evolve under increasing strain. The held-out testing file (file 20) validates generalisation to unseen strain conditions, simulating a real extrapolation scenario relevant to yield strength, ductility, and hardness prediction.

4. Experimental Results

All four LSTM variants are evaluated on Euler angle prediction across 20 strain conditions. RMSE and Inverse Pole Figure (IPF) maps serve as primary evaluation criteria.

4.1 RMSE Comparison

Model	RMSE	Notes
Deterministic LSTM	5.4855	Best accuracy; no uncertainty output
Last-Layer SVI (Partial)	13.3633	Good balance of accuracy and UQ
Last-Layer SGLD (Partial)	14.7996	Competitive; stable training
Fully Stochastic SVI	19.3111	Highest RMSE; most expensive

Table 1: RMSE comparison across model variants (lower is better). Highlighted row indicates best performance.

4.2 Key Findings

- The **deterministic model achieves the lowest RMSE (5.49)**, confirming its advantage in pure predictive accuracy on this regression task.
- **Partial stochasticity (last-layer SVI, RMSE 13.36) outperforms full stochasticity (RMSE 19.31)** — directly challenging the assumption that more Bayesian parameters improve performance.
- The **SGLD variant (RMSE 14.80)** offers competitive uncertainty modelling with more stable training dynamics than full SVI.

- Both stochastic models produce **more structured, dispersed error patterns** in IPF maps compared to the deterministic baseline, indicating improved uncertainty capture even at higher RMSE.
 - The deterministic model shows **concentrated error in specific spatial regions**, revealing limited capacity to represent predictive uncertainty.
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5. Lessons Learned

Selective vs. Full Stochasticity

Full stochasticity across all model parameters is not necessary for effective uncertainty estimation. Output-layer stochasticity alone achieves comparable uncertainty calibration with substantially lower computational cost — a practically significant finding for real-world Bayesian deep learning deployments.

Uncertainty Estimation in Scientific Prediction

While LSTMs excel at learning temporal dependencies, uncertainty quantification is critical in materials science and other safety-critical domains. Selective Bayesian methods provide a pragmatic path toward uncertainty-aware predictions without sacrificing tractability.

Challenges with Hamiltonian Monte Carlo

Initial attempts to use HMC encountered vanishing step sizes and low acceptance probabilities, making it impractical at our model scale. Switching to mean-field VI and SGLD provided stable, scalable alternatives for posterior approximation.

6. Future Work

- Extend stochasticity to **hidden recurrent layers** to capture uncertainty in intermediate temporal states and improve generalisation to unseen conditions.
 - Compare **ELBO scores** across all model variants for a principled assessment of posterior approximation quality.
 - Evaluate models on **real-world outdoor microstructure data** to assess robustness beyond the synthetic benchmark.
 - Incorporate a **passive texture/depth anti-spoofing CNN** within remaining model budget to harden uncertainty estimates against distributional shift.
 - Explore **hardware acceleration (NNAPI / Core ML)** to reduce inference latency for deployment on edge devices.
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7. Conclusion

This project investigated Bayesian LSTM models for sequence modelling and uncertainty estimation, specifically predicting Euler angle evolution in synthetic microstructure data under incremental strain. Four model variants spanning deterministic, partially stochastic, and fully stochastic designs were evaluated

empirically.

Our findings challenge the prevailing assumption that full stochasticity is required for principled uncertainty estimation. **Partially stochastic models — with stochasticity limited to the output layer — provide comparable or superior performance at a fraction of the computational cost.** The accuracy ranking from best to worst is: Deterministic (RMSE 5.49) > Partial SVI (13.36) > Partial SGLD (14.80) > Fully Stochastic SVI (19.31).

These results suggest that targeted stochasticity offers a practically superior trade-off for real-world scientific prediction tasks that require both accuracy and calibrated uncertainty estimation.

References

- [1] Sharma, A. et al. (2023). *Do Bayesian Neural Networks Need To Be Fully Stochastic?* arXiv:2211.06291.
- [2] Project Repository: github.com/Meetlalwani01/CS772-